

-- Cox Regression - Sarcopenia Onset -----

i Definition : fnih
i Covariates : fixed
i Dairy : continuous
i Route : mice
i Interaction: none

-- MICE route - pooling across imputed datasets --

Detected 20 imputed datasets.
Fitting .imp = 1 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 2 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 3 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 6 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 7 / 20 ...
Events: 36 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 34 events
Fitting .imp = 8 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 9 / 20 ...
Events: 36 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 34 events
Fitting .imp = 10 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 12 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 13 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 14 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 32 events
Fitting .imp = 15 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1117 rows | 33 events
Fitting .imp = 16 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

```
Fitting .imp = 18 / 20 ...  
Events: 35 sarcopenia cases out of 1117 participants
```

```
-- Fixed covariates dataset (one row per pt) --
```

```
??? Fixed dataset: 1117 rows | 33 events  
Fitting .imp = 19 / 20 ...  
Events: 34 sarcopenia cases out of 1117 participants
```

```
-- Fixed covariates dataset (one row per pt) --
```

```
??? Fixed dataset: 1117 rows | 32 events  
Fitting .imp = 20 / 20 ...  
Events: 34 sarcopenia cases out of 1117 participants
```

```
-- Fixed covariates dataset (one row per pt) --
```

```
??? Fixed dataset: 1117 rows | 32 events
```

```
-- Pooled adjusted model results (Rubin's rules) --
```

```
# A tibble: 8 x 5 term HR CI_low CI_high p.value <chr> <dbl> <dbl> <dbl> <dbl>  
1 dairy_exposure 0.999 0.997 1.00 0.437 2 BMI 1.16 1.07 1.25 0.000829 3  
education_levelMedium (ISCED 3-4) 0.652 0.299 1.42 0.267 4 education_levelHigh  
(ISCED 5-8) 0.0000000116 0 Inf 0.997 5 smoking_statusFormer 1.09 0.506 2.36  
0.813 6 smoking_statusCurrent 0.950 0.200 4.51 0.946 7 mvpa_min_day_f1 1.00  
0.997 1.00 0.760 8 diabetes_statusDiabetes 0.511 0.135 1.94 0.308
```

-- 1. Proportional Hazards - cox.zph() | pass: global p > 0.05

-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear

'geom_smooth()' using formula = 'y ~ x'

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Linearity plots saved to

03_outputs/Cox/mice_fnih_fixed_continuous_27072026_1812

-- 3. Outliers - deviance & martingale residuals | pass: |dev| <= 3

i 2 observation(s) with |deviance residual| > 3.

-- 4. Influential observations - std. DFBetas | pass: no extreme spikes

i 21 obs exceed |DFBeta_std| > 0.2.

-- 5. Overall Fit - Cox-Snell residuals | pass: points on 45° diagonal

Cox-Snell plot saved to 03_outputs/Cox/mice_fnih_fixed_continuous_27072026_1812

-- 6. Multicollinearity - VIF | pass: VIF < 5 (< 10 relaxed)

GVIF Df GVIF^(1/(2*Df)) dairy_exposure 1.020 1 1.010 BMI 1.117 1 1.057

education_level 1.016 2 1.004 smoking_status 1.034 2 1.008 mvpa_min_day_f1

1.035 1 1.017 diabetes_status 1.110 1 1.054

??? No collinearity concerns (all VIF/GVIF within range).

-- 7. Log-log survival plot | pass: parallel lines across log(time)

```
quartile n_pts n_events person_years rate_per_1000py ci_low ci_high 1 Q1 (low)
280 7.3 19850.00 0.37 0.15 0.75 2 Q2 279 8.0 19759.30 0.40 0.17 0.80 3 Q3 279
9.3 19831.52 0.47 0.22 0.88 4 Q4 (high) 279 8.0 19858.90 0.40 0.17 0.79
Incidence table saved to
03_outputs/Cox/mice_fnih_fixed_continuous_27072026_1812
Model results saved to
03_outputs/Cox/mice_fnih_fixed_continuous_27072026_1812/mice_model_results.csv
```